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AI for Bharat Hackathon

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Team Name : The Sentinels

Team Leader Name : Abhinav Khare

Problem Statement : Build an AI-powered solution that helps people learn faster, work smarter, or become more productive while building or understanding technology.

Brief about the Idea:

Code Guardian is an educational VS Code extension designed to intercept AI-generated code. Instead of blindly pasting, developers must explain the code logic in their own words. Our AI backend grades their understanding. If they fail three times, it switches to a supportive "Explain Like I'm 5" (ELI5) mentor mode.

Why AI is required: Traditional rule-based systems cannot evaluate if a student genuinely understands code. We need an LLM to semantically grade natural language explanations.

How AWS services are used: AWS Lambda runs the serverless Python backend to orchestrate the AI grading and fallback routing. Amazon DynamoDB persistently logs user session data, scores, and strike counts.

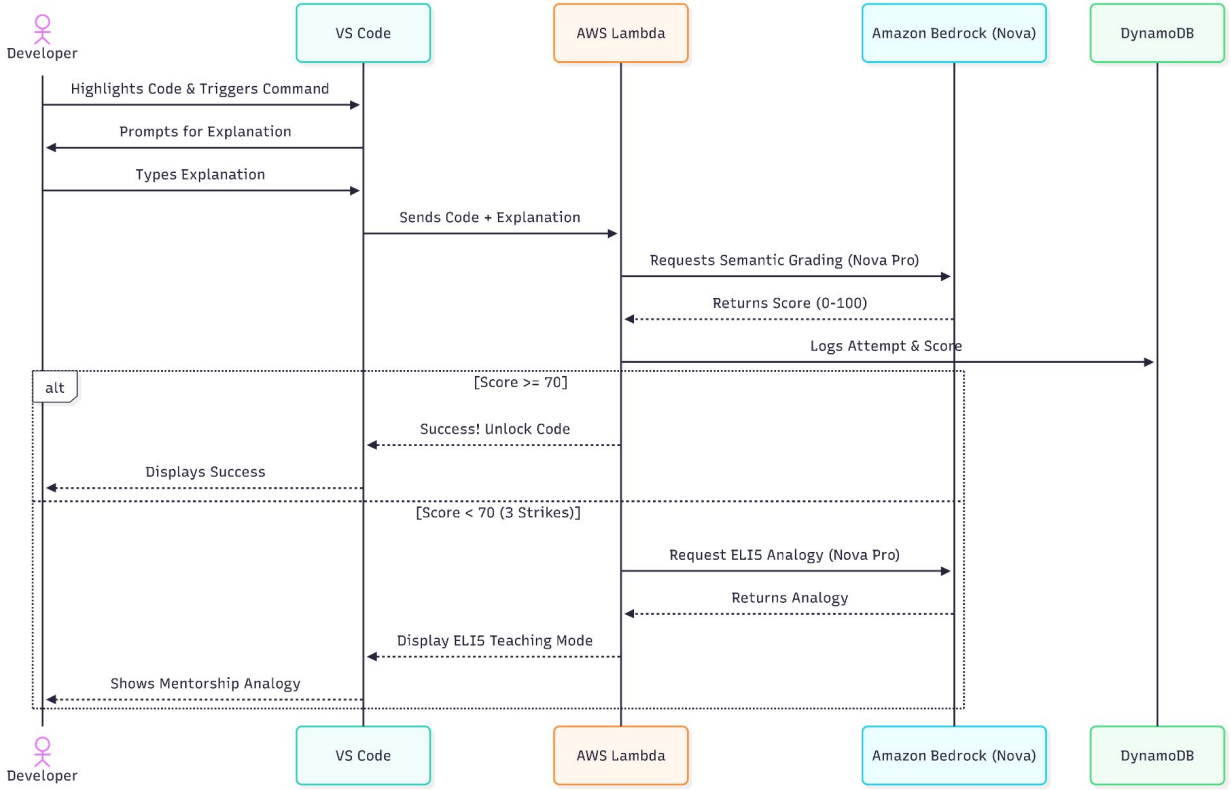
Value the AI layer adds: It transforms a "friction point" into a personalized educational mentor, preventing skill degradation.

The AI Engine: Amazon Bedrock provides the state-of-the-art generative AI capabilities required to semantically evaluate developer explanations.

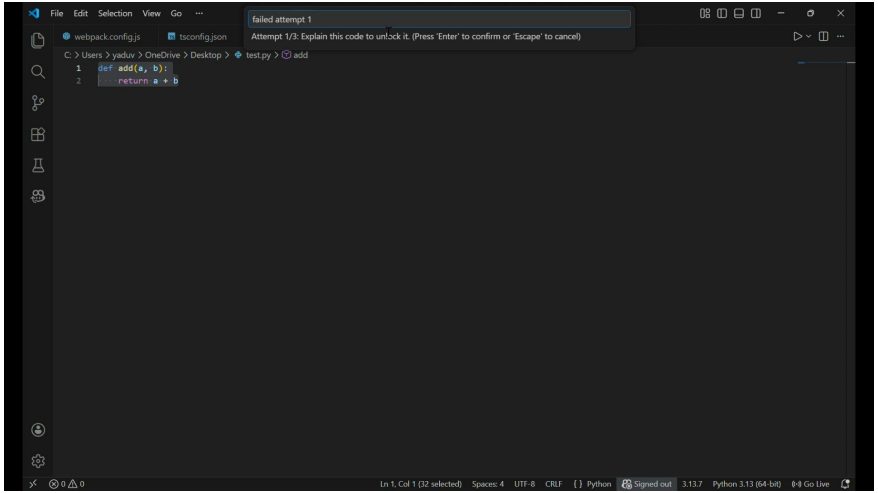
List of features offered by the solution:

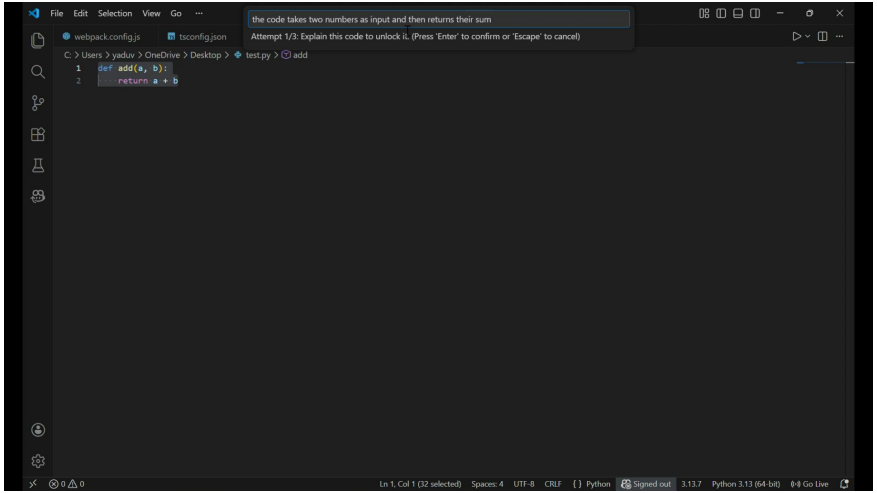
- **Real-Time Semantic Grading:** Scores explanations on a 0-100 scale.
- **The 70% Threshold:** Unlocks the developer's workflow only if they prove understanding.
- **"3-Strike" ELI5 Fallback:** Automatically simplifies complex logic using real-world analogies after 3 failed attempts.
- **Cloud-Based Tracking:** Logs all learning history in DynamoDB.
- **Enterprise Fallback System:** Built-in fault tolerance automatically reroutes requests to a backup Bedrock model (Nova Micro) if the primary model times out, ensuring zero downtime.

Process flow diagram or Use-case diagram

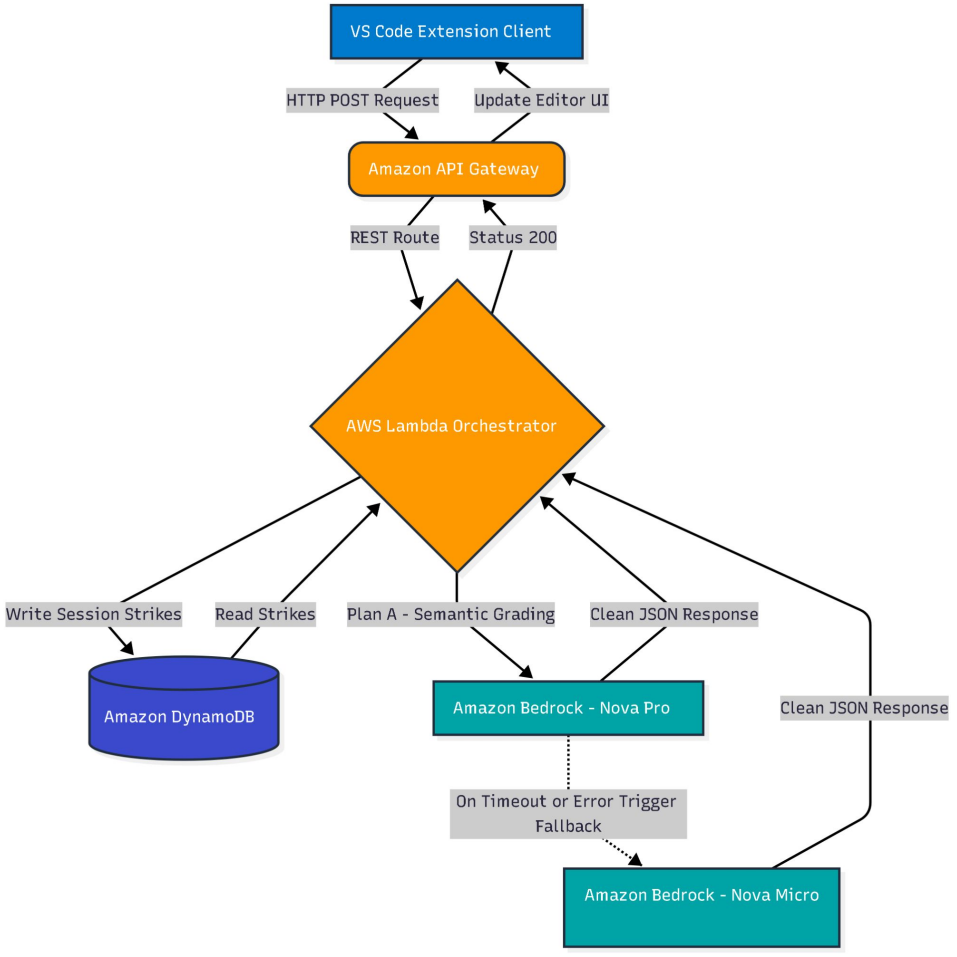


Wireframes/Mock diagrams of the proposed solution





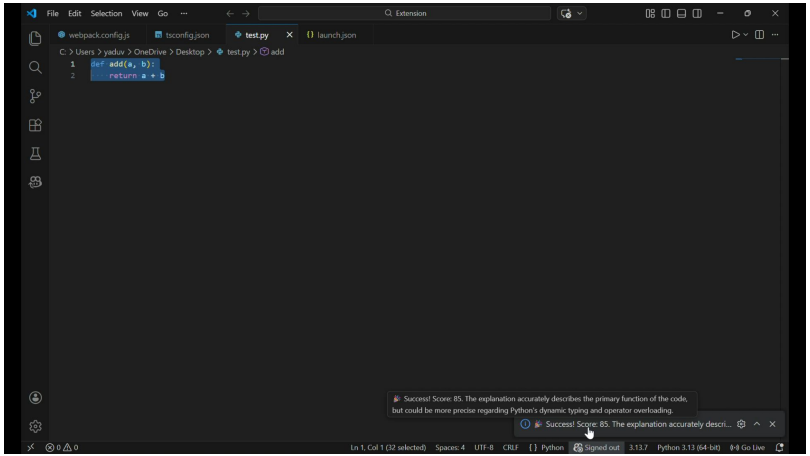
Architecture diagram of the proposed solution:

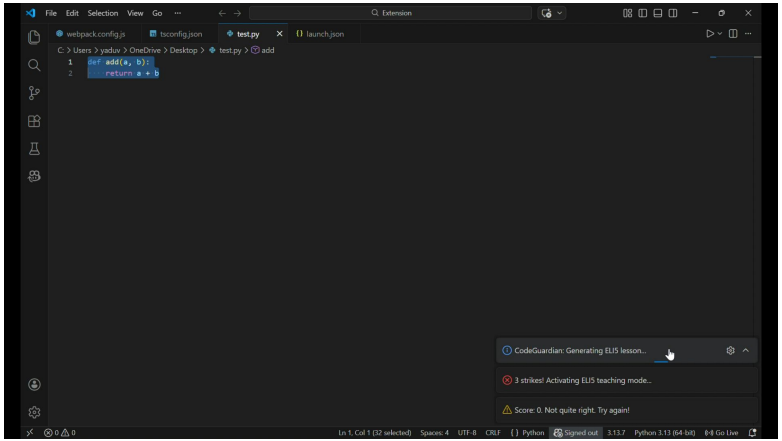


Technologies utilized in the solution:

- **Frontend:** Visual Studio Code Extension API, TypeScript, Node.js
- **Backend Cloud:** AWS Lambda, Amazon API Gateway
- **Database:** Amazon DynamoDB
- **AI/LLM Engine:** Amazon Bedrock
- **Primary Model:** Amazon nova-pro-v1:0 (Handles complex semantic grading and ELI5 analogies).
- **Enterprise Resilience:** Built-in AWS Lambda fallback routing to Amazon nova-micro-v1:0 to guarantee uninterrupted service during latency spikes.

Snapshots of the prototype:





Prototype Performance report/Benchmarking:

Latency: Sub-2 second end-to-end response time from VS Code to AWS Lambda and back.

Uptime: High availability powered by AWS Serverless architecture.

Accuracy: Effectively evaluates logic using semantic meaning rather than basic keyword matching.

Fault Tolerance: Automated multi-model fallback routing ensures continuous availability, eliminating the single point of failure common in standard AI integrations.

Additional Details/Future Development:

- **Auto-Paste Interception:** Hooking directly into VS Code's clipboard API (moving beyond the MVP highlight requirement).
- **Multilingual Support:** Explanations and ELI5 analogies in regional languages (Hindi, Tamil, etc.).
- **Voice-to-Explain:** Audio transcription for verbal explanations.
- **Gamification:** Earning knowledge points and daily streaks.
- **Interview Prep Mode:** Dynamically generated Q&A rounds based on pasted code.
- **Code Optimization:** Contextual AI tips (e.g., time complexity, library fun facts).

Prototype Assets:

- ➔ GitHub Public Repository Link:
- ➔ https://github.com/yaduvansh-kumra/CodeGuardian---AI-Hackathon
- ➔
- ➔ Demo Video Link :
  https://drive.google.com/drive/folders/1ojrNcGFhb4a3t9hZV0OZtstvsjZ9sqKD?usp=sharing

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Thank You